Engineering Calculus
EMAT101L

Lecture 3

School of Engineering Science & Technology,

Bennett University

September 14, 2023

Supremum & Infimum

Let $S(\neq \phi) \subseteq \mathbb{R}$.

Supremum (or least upper bound)

If S is bounded above, then the **smallest number** K **among all the upper bounds** of S is said to be the supremum of S. Denoted by $sup(S)$.

Infimum (or greatest lower bound)

If S is bounded below, then the **largest number** k **among all the lower bounds** of S is said to be the infimum of S. Denoted by $inf(S)$.

Example: $S = [1, 3, 5, 7, 9]$.
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Examples

Q1: $S = \{x : x > 0, x \in \mathbb{R}\}$.Find $inf(S)$, and $sup(S)$.

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Ans: $inf(S) = 0$ and $sup(s)$ do not exist. Also, infimum do not belong to the set.

Q2: The infinite set $S = \{x : 0 \leq x \leq 1, x \in \mathbb{Q}\}$. Is this set bounded?

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Q2: The infinite set $S = \{x : 0 \leq x \leq 1, x \in \mathbb{Q}\}$. Is this set bounded?
Ans: The set S is bounded.
$sup(S) = 1$ and $inf(S) = 0$, both belongs to the set.

Q3: Consider $S = \{\frac{1}{n} : n \in \mathbb{N}\}$. Is this bounded?

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Ans: The set S is bounded. $sup(S) = 1$, $sup(S) \in S$.

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Ans: The set S is bounded. $sup(S) = 1$, $sup(S) \in S$.
$inf(S) = 0$, $\inf(S) \notin S$.

Q4: Discuss the bounds of: (i) [a,b], (ii) (a,b], (iii) [a,b), (iv) (a,b).

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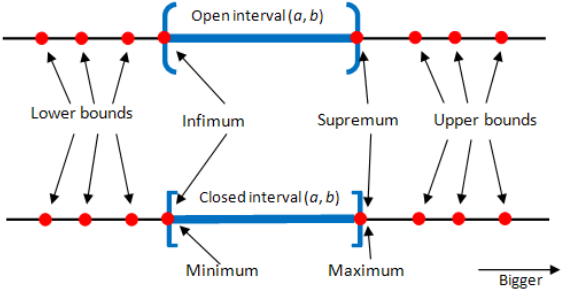
Maxima/Minima, Lower/Upper bounds, Supremum/Infimum

Note 1: Upper bound and Lower bound **may not belong** to the set.

Note 2: Maximum and minimum **always belong** to the set.

Note 3: Supremum and Infimum **may not belong** to the set.



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